

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

ME-7004(1)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Machine Tool Design

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What is the importance of Machine tools?
b) What are different motions provided on a lathe?
2. a) Discuss the selection of number of teeth on a milling cutter.
b) What is machine tool? Classify machine tools according to (i) Function of the machine and (ii) The geometrical shapes of the work pieces to be machined.
3. a) Describe speed stepping according to arithmetic progression.
b) A gear box for a drilling machine is to give six spindle speeds. If the size of the drills to be used is from 5 to 20 mm diameter and the cutting speed is 21m/min. Calculate the common ratio for G.P. Series of speeds and hence the various spindle speeds.
4. What are Feed control devices? Why are they needed? Sketch and explain any control device in detail.

ME-7004(1)-CBGS

PTO

[2]

5. a) What is chip formation? Name the different types of chips formed in metal cutting. Describe each type with the help of neat sketches.
- b) During orthogonal turning operation the following data was obtained:
Cutting force = 120 kg, feed force = 30 kg
Rake angle = 10° , feed = 0.2 mm/revolution
Width of cut = 2.3 mm, chip thickness = 0.4 mm
Cutting speed = 120 m/min
Calculate the following:
i) Chip thickness ratio
ii) Shear angle
iii) Shear stress
6. A 4 speed (2×2) gear box is required to be designed for transmitting 8 H.P. with speed ranging from 300 RPM with $\phi = 1.2$ select an optimum ray diagram and hence calculate gear sizes module and width of gears.
7. Why is it necessary to broach internal helical gear when the helix angle is greater than 15 degrees?
8. Write short notes : (any two)
- a) Bed and column sections in machine tools
 - b) Thick and thin film lubrication
 - c) Feed mechanism used in milling machine
 - d) Name the main elements of a lathe machine
